

Backend Foundation Summary

Blood Donation and Donor Management System

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Role	Backend Foundation / Core Setup
Completed	Database connection, common enums, JPA entities, relationships, repositories, and initial backend run test

Overview

Waxaan diyaariyey backend foundation-ka project-ka Blood Donation and Donor Management System. Waxaan sameeyey PostgreSQL database la yiraahdo blood_donation_db, kadib waxaan backend-ka ku xiray database-ka. App-ka si sax ah ayuu ugu run-gareeyay localhost:8080.

Common Enums

Waxaan sameeyey common/enums folder-ka, waxaana ku daray enums-ka system-ka isticmaali doono: UserRole, UserStatus, Gender, DonorStatus, InventoryStatus, DonationStatus, AnnouncementStatus, iyo UrgencyLevel.

Entity Layer

Waxaan sameeyey entity layer-ka matalaya database tables-ka: User, BloodCategory, Donor, Inventory, Donation, iyo Announcement. Entity-yada waxaa lagu isticmaalay JPA annotations sida @Entity, @Table, @Id, @GeneratedValue, @Column, @OneToOne, @ManyToOne, iyo @JoinColumn.

Database Relationships

Waxaan sameeyey relationships-ka muhiimka ah: User iyo Donor, BloodCategory iyo Donor, BloodCategory iyo Inventory, Donor iyo Donation, BloodCategory iyo Donation, iyo BloodCategory iyo Announcement.

Repository Layer

Waxaan sameeyey repositories-ka: UserRepository, CategoryRepository, DonorRepository, InventoryRepository, DonationRepository, iyo AnnouncementRepository. Repositories-kan waxay noo oggolaanayaan methods diyaar ah sida save(), findById(), findAll(), deleteById(), existsById(), iyo count(). Waxaan sidoo kale ku daray custom methods sida existsByEmail, findByEmail, findByStatus, findByCategoryId, findByDonorId, iyo findByDonationDateBetween.

Backend Run Test

Waxaan run-gareeyay backend-ka, wuxuuna si sax ah u shaqeeyay. Console-ku wuxuu muujiyay in PostgreSQL connection-ku successful yahay, JPA repositories la helay, Hibernate initialized yahay, Tomcat-na ka shaqeynayo port 8080.

Summary

Waxaan dhisay saldhigga backend-ka: database connection, common enums, JPA entities, database relationships, repositories, iyo initial backend run test. Qaybtan waxay diyaar u tahay in lagu sii dhiso service layer, controller layer, authentication, JWT security, iyo modules-ka kale.

Next Steps

Waxa xiga waa exception handling, Auth service, JWT security, Category service/controller, iyo integration-ka modules-ka kale sida Donor, Inventory, Donation, Announcement, Dashboard, iyo Report.